

Analysis group meeting

Mathieu Ouillon
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MISSISSIPPI STATE
UNIVERSITY™

Cooking chef

- Timeline pass0v6 end last Thursday (very fast, only two day).
- Cooking for 2nd round for the FTOF/CTOF and CND calibration:
 - 3 workflows all at 99%

Name	Complete	Jobs	Undispatched	Dispatched	Preparing	Pending	Running	Reaping	Other	Problems
rgd-dmra-calib_v9_1-18321x9	99.93%	3033	1	1	0	0	1	0	0	0
rgd-dmra-calib_v9_3-18312x2	99.75%	789	2	0	0	0	0	0	0	0
rgd-dmra-calib_v9_2-18809x10	99.84%	3829	3	3	0	0	3	0	0	0

- Skimming the pass0v6, can be found here:
/cache/hallb/scratch/rg-d/production/skim_pass0v6/

Nuclear transparency

- Carbon data:
 - path: /cache/hallb/scratch/rg-d/production/Bspot/v5dstCxC/dst/recon/
- run list:
 - 018339/ 018341/ 018343/ 018346/ 018441/ 018443/ 018445/ 018498/
018756/ 018340/ 018342/ 018344/ 018440/ 018442/ 018444/ 018475/
018524/ 018850/
- LD2 data:
 - path: /cache/hallb/scratch/rg-d/production/Bspot/v5dstLD2/dst/recon/
- run list:
 - 018421, 018429, 018439, 018656, 019058, 018424, 018431, 018528,
018851, 018427, 018432, 018559, 018873, 018419, 018428, 018433,
018644, 019021
- Only outbending runs.

Nuclear transparency

- electron :
 - `REC::Particle::pid == 11` and `REC::Particle::status < 0`
 - $-12 < v_{z_e} < 5$ cm
 - $-5 < \text{REC::Particle::chi2pid} < 5$
- π^- :
 - $-10 < \text{REC::Particle::chi2pid} < 10$
- π^+ :
 - $-10 < \text{REC::Particle::chi2pid} < 10$
- invariant:
 - $W > 2$ GeV
 - $z_h > 0.9$
 - $0.1 < -t < 0.5$ GeV²
 - $l_c < 1$ fm

Nuclear transparency

- The Breit-Wigner function for the fit of the invariant mass two pions:

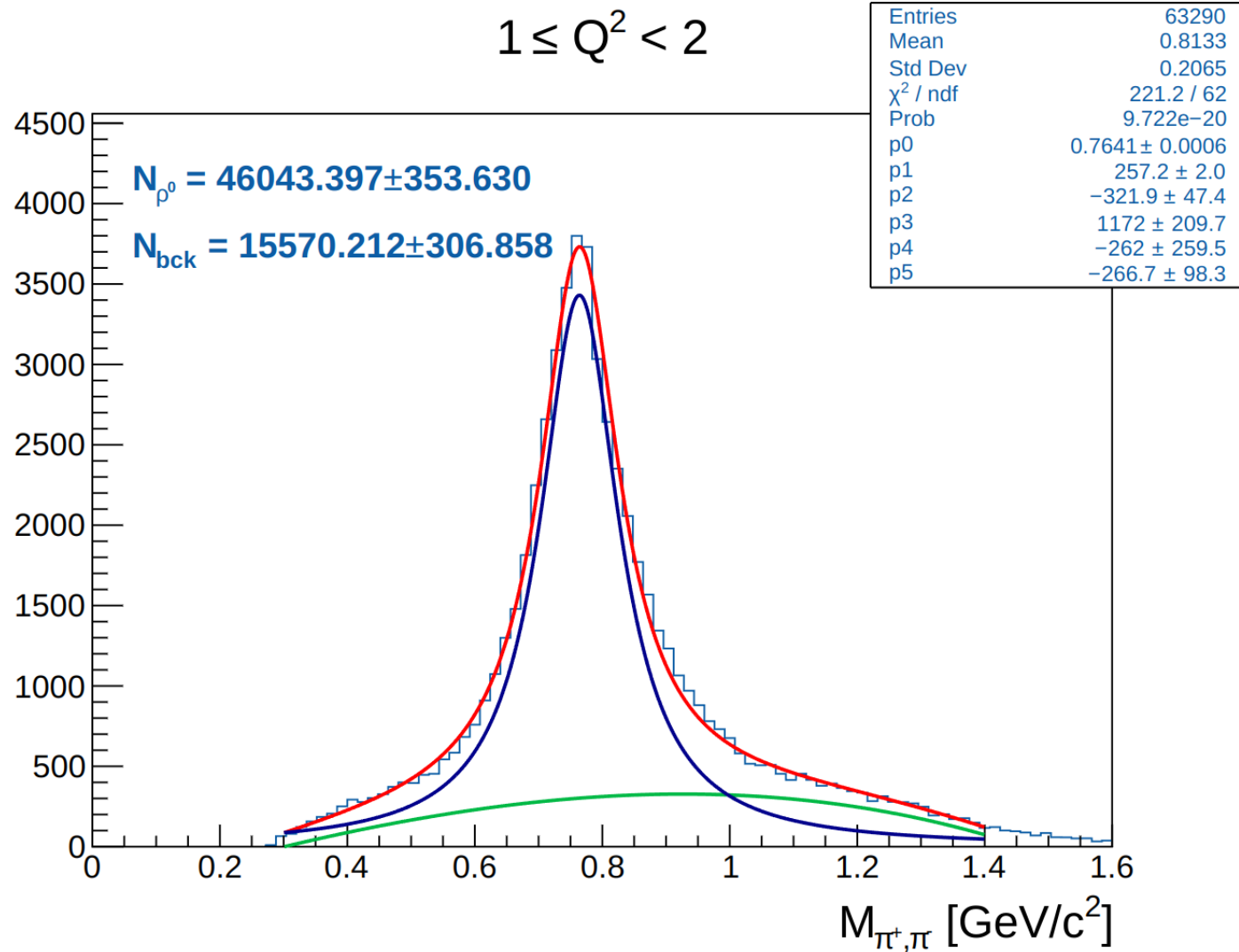
$$BW(M_{\pi^+\pi^-}; \alpha, \beta) = \beta \frac{\Gamma}{(M_{\pi^+\pi^-} - \alpha)^2 + \Gamma^2}$$

- The polynomial function

$$pol3(M_{\pi^+\pi^-}; a, b, c, d) = a + bM_{\pi^+\pi^-} + cM_{\pi^+\pi^-}^2 + dM_{\pi^+\pi^-}^3$$

- The yield for ρ^0 is computed with the integral of the BW function
 - Use ROOT function
 - And test [Simpson's rule](#)
 - Give the same result.

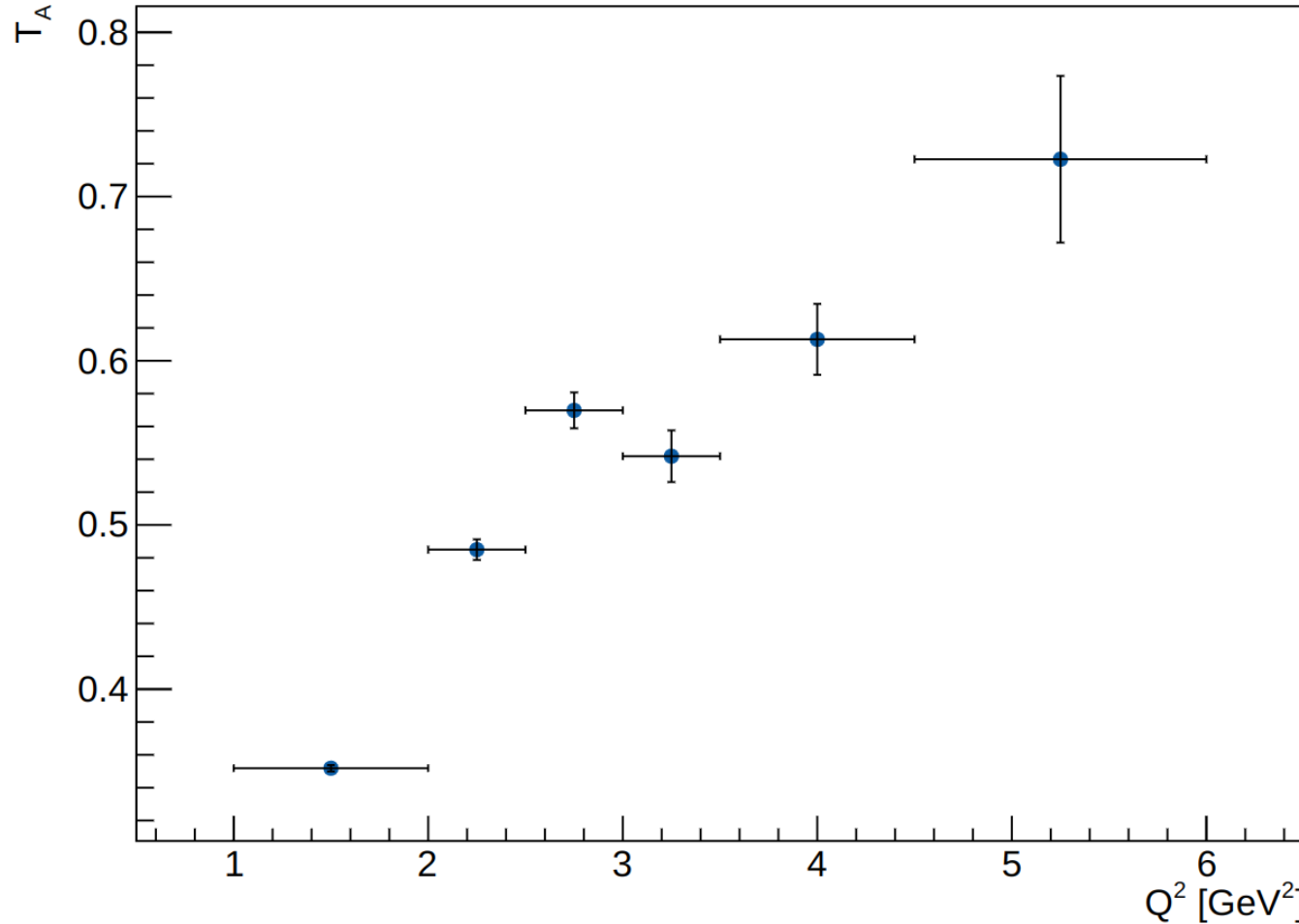
Nuclear transparency



Carbon data.

Test different implementations for the fit function:

Nuclear transparency



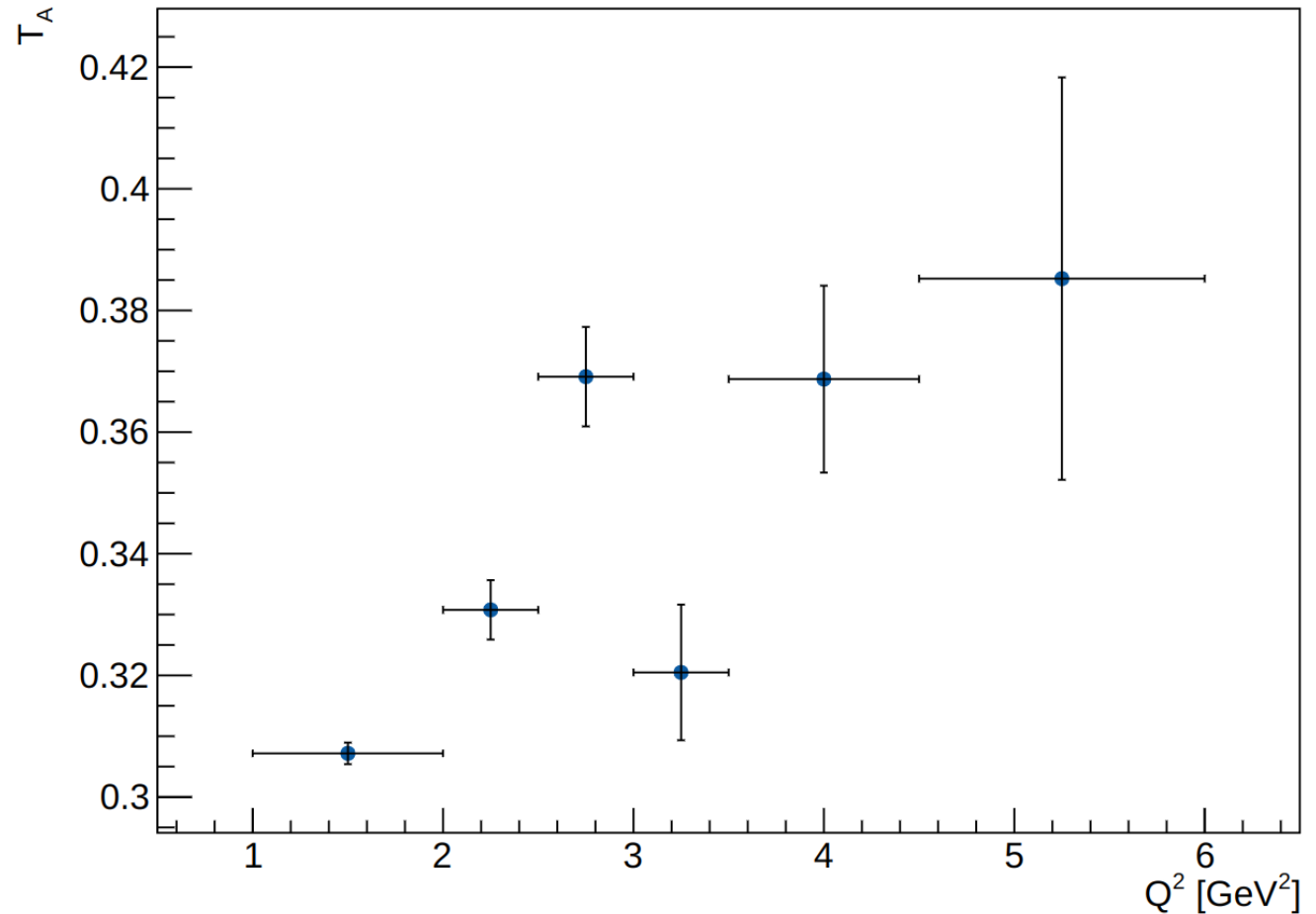
- The nuclear transparency is computed with:

$$T_A = \frac{N_C}{N_D} \frac{\rho_D r_D}{\rho_C r_C}$$

with $\rho_D = 0.164 \text{ g}\cdot\text{cm}^{-3}$, $r_D = 5 \text{ cm}$, $\rho_C = 2.2 \text{ g}\cdot\text{cm}^{-3}$ and $r_C = 0.4 \text{ cm}$. N_C is the yield of ρ^0 for the carbon target. N_D is the yield of ρ^0 for the liquid deuterium target.

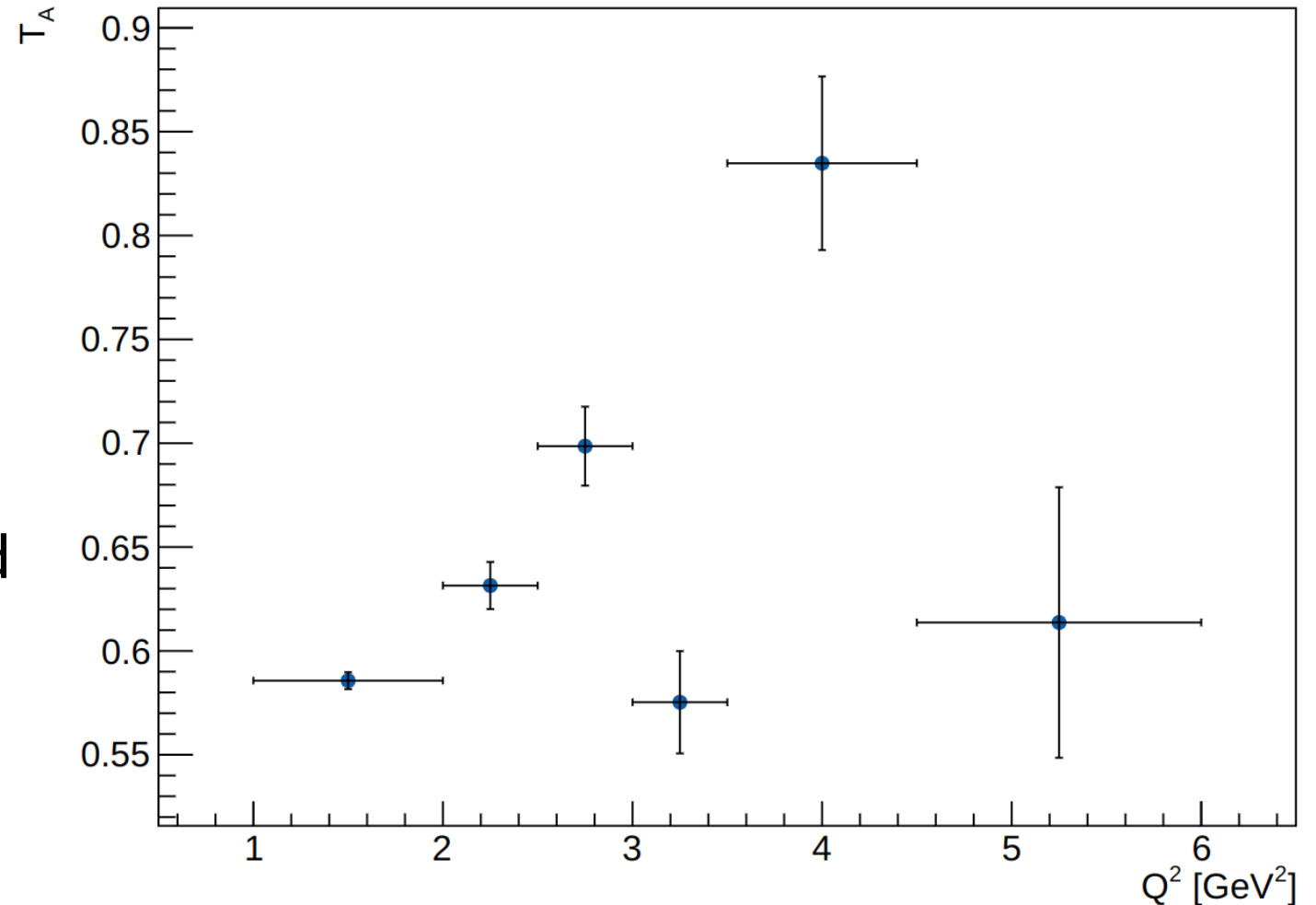
Nuclear transparency

- 1st: Use only outbending for both C and LD2:
- 018440, 018441, 018442, 018443, 018444, 018445, 018475, 018498, 018524, 018756, 018850
- Overall lower, so the number of events change the nuclear transparency



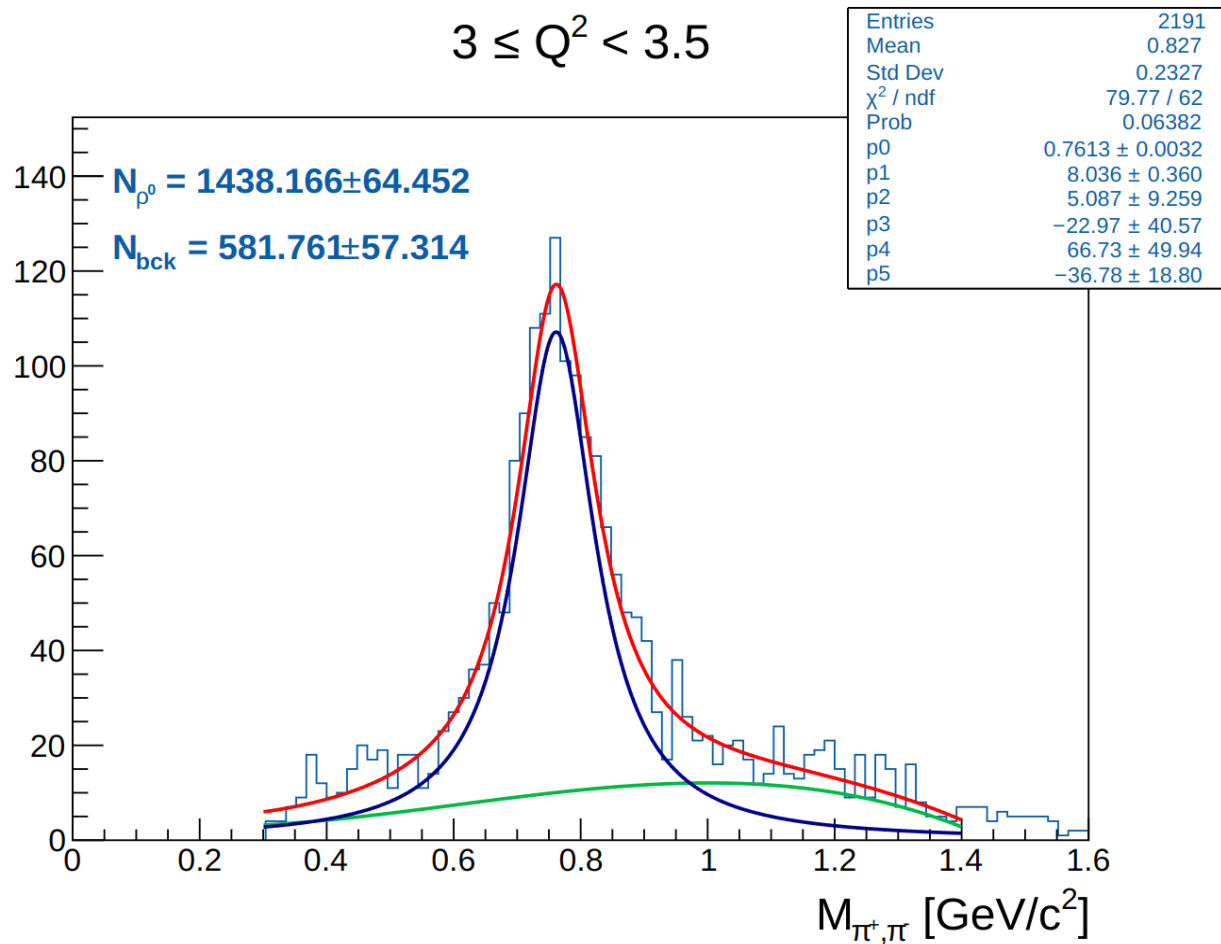
Nuclear transparency

- Use same amount of event in C and LD2:
 - The number of total event processed is 200 M.
- Better agreement with the proposal
- Issues with the bin 3.25 and 5.25.

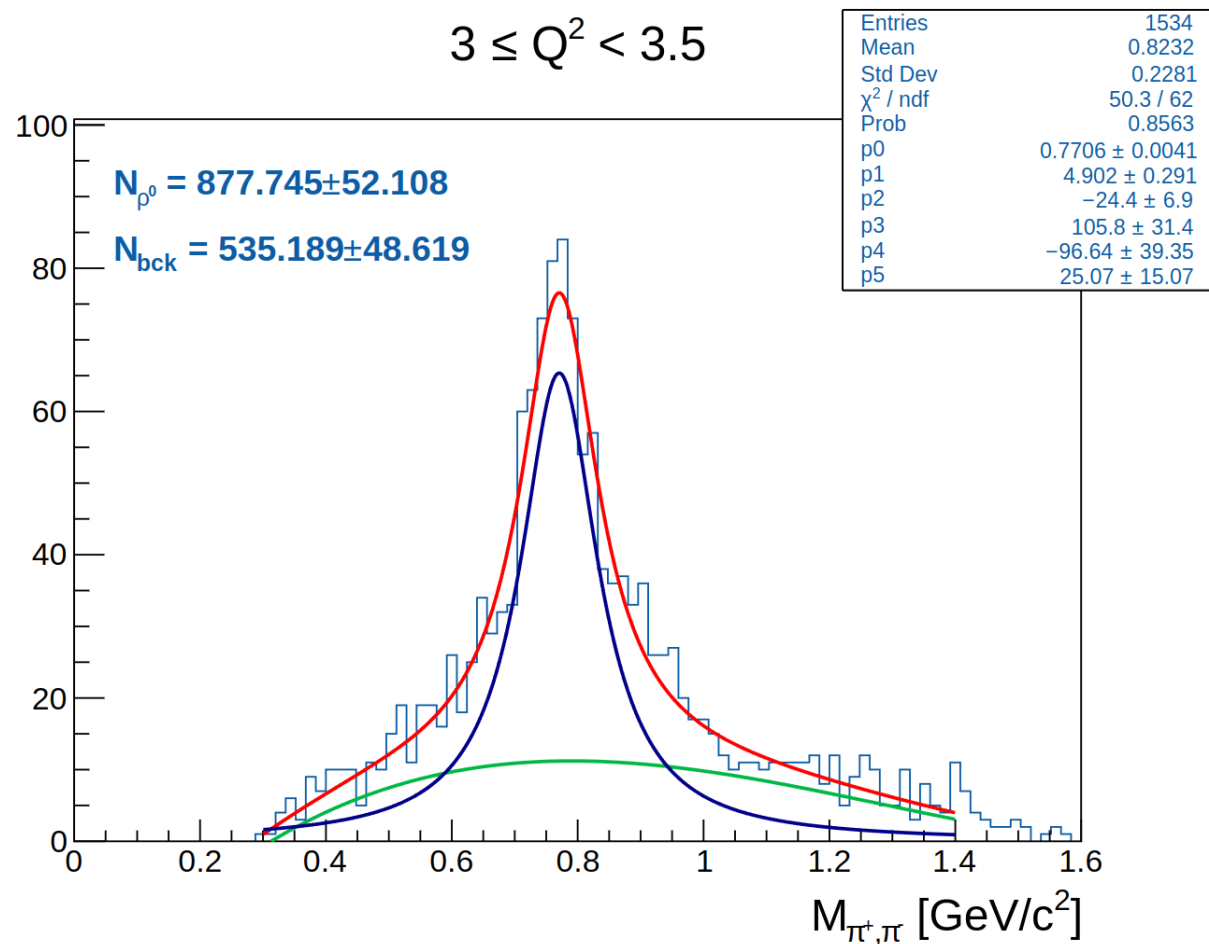


Nuclear transparency

LD2



Carbon



Nuclear transparency

- Compute the nuclear transparency using the number of events processed:

$$T_A = \frac{N_C}{N_D} \frac{\rho_D r_D N_D^{\text{event}}}{\rho_C r_C N_C^{\text{event}}}$$

The nuclear transparency is:

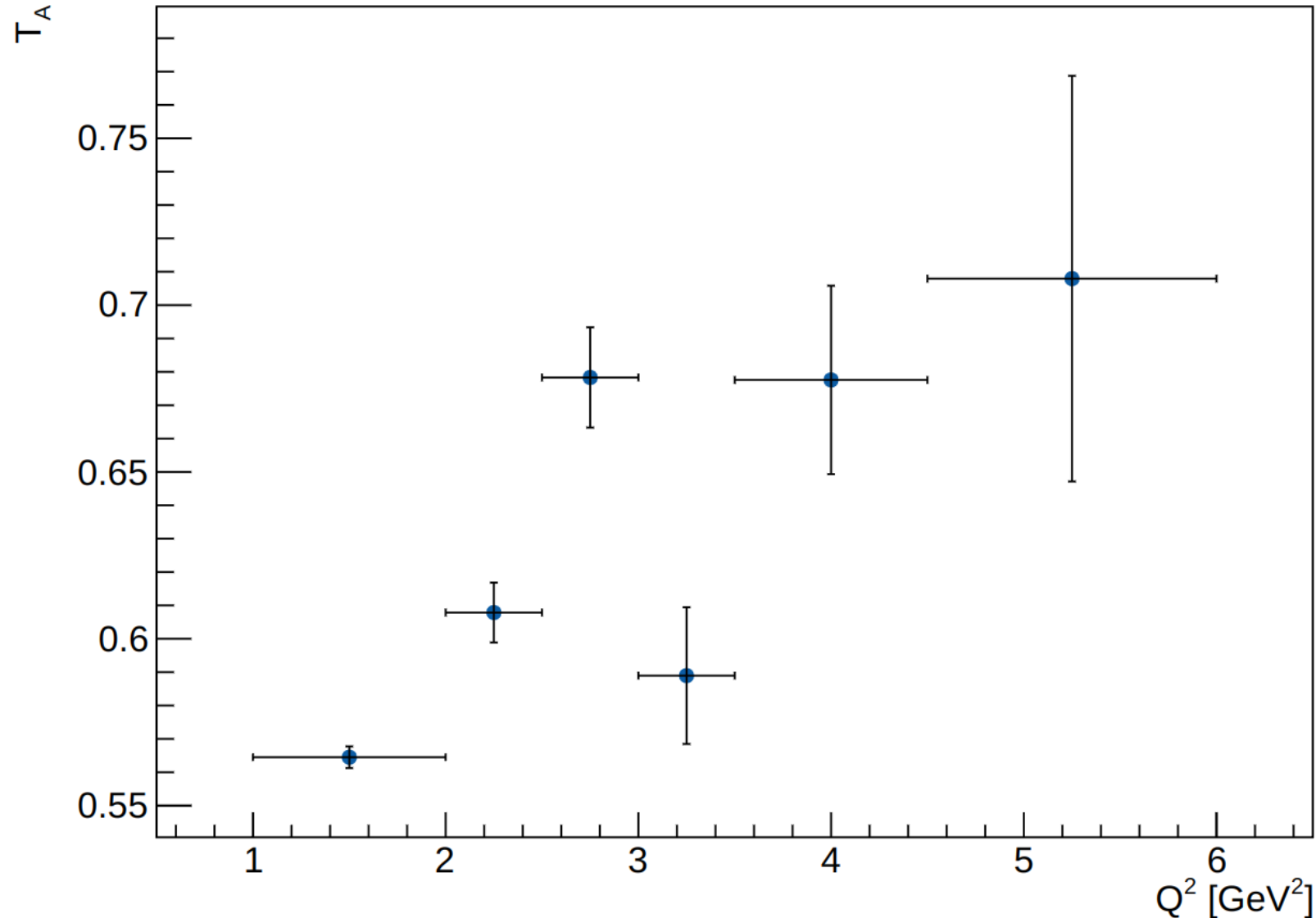
$$T_A = \frac{N_C}{N_D} \frac{L_D^{\text{int}}}{L_C^{\text{int}}}$$

with L^{int} being the integrated luminosity:

$$L^{\text{int}} = \rho r N_{Av} \frac{Q^{\text{int}}}{q_e}$$

So in this case the nuclear transparency become:

$$T_A = \frac{N_C}{N_D} \frac{\rho_D r_D Q_{\text{int}}^D}{\rho_C r_C Q_{\text{int}}^C}$$

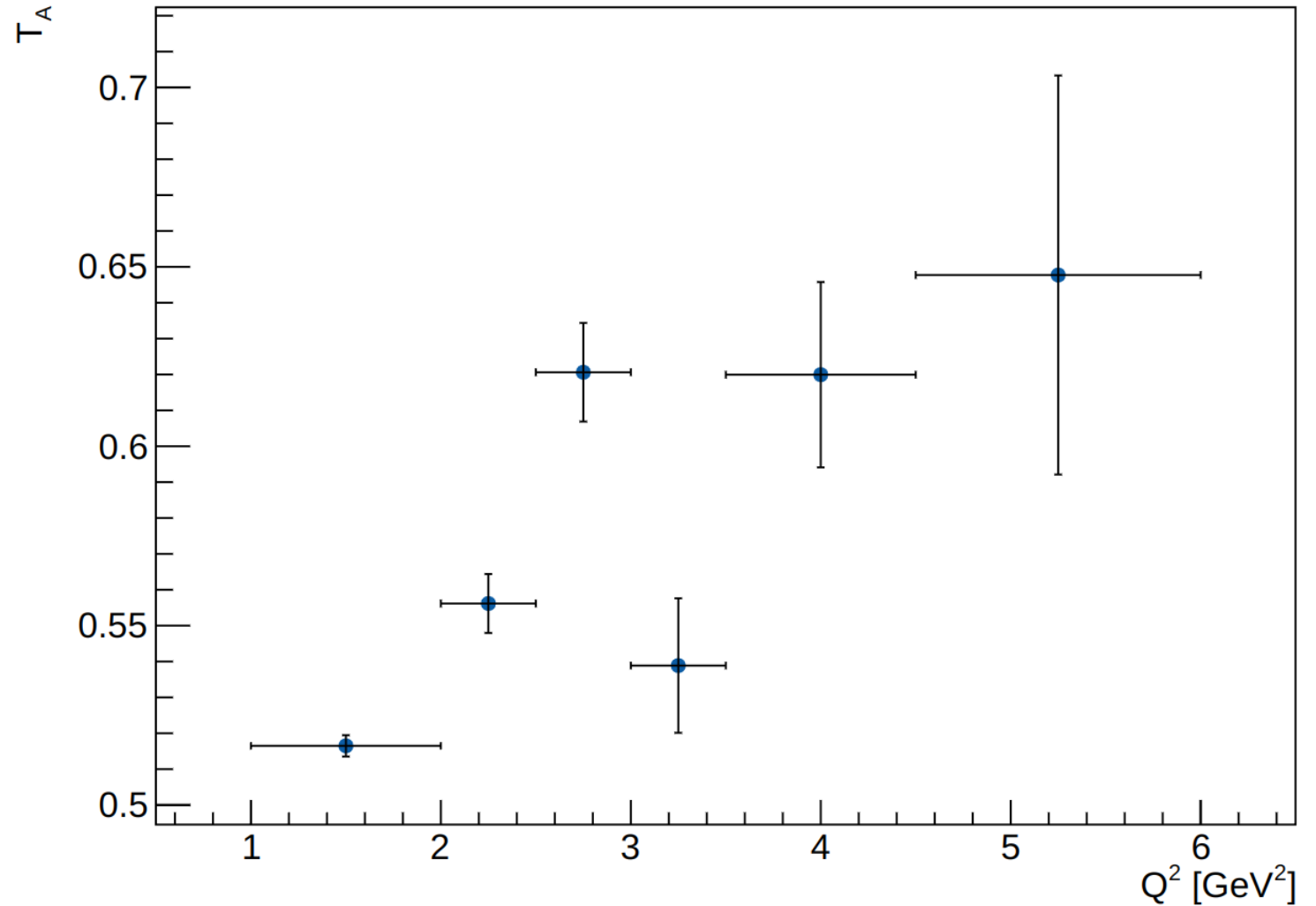


Nuclear transparency

- Using the beam charge for the nuclear trans

$$T_A = \frac{N_C}{N_D} \frac{\rho_D r_D Q_{int}^D}{\rho_C r_C Q_{int}^C}$$

- Overall smaller than the previous one.

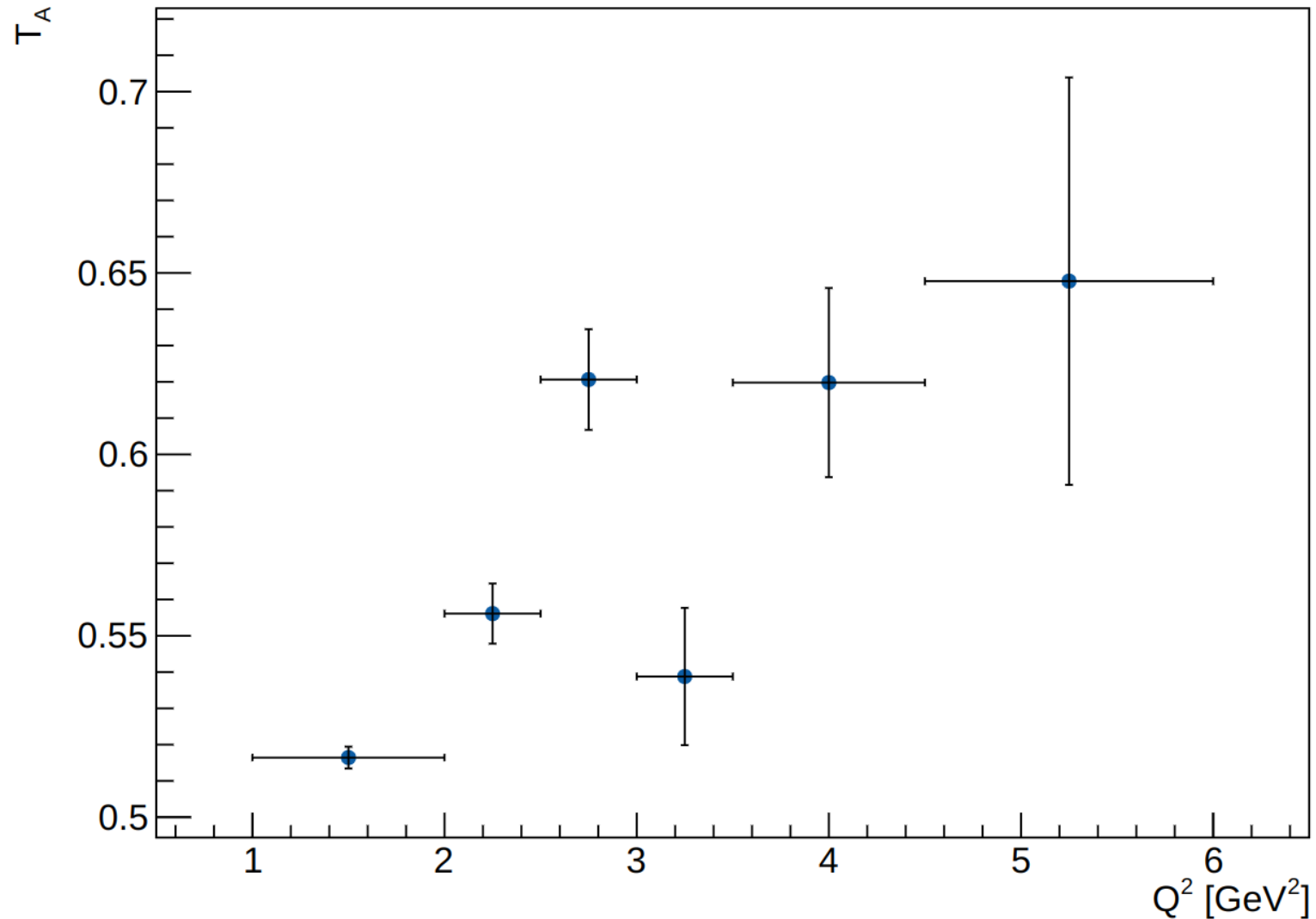


Nuclear transparency

- How to get the beam charge:
 - I use this presentation :
https://clasweb.jlab.org/wiki/images/2/26/RGD_April_24_2024.pdf
 - Run the python script and notebook
 - End up with a file with the beam charge in mC for each runs.
- <https://github.com/mholtrop/RunData>

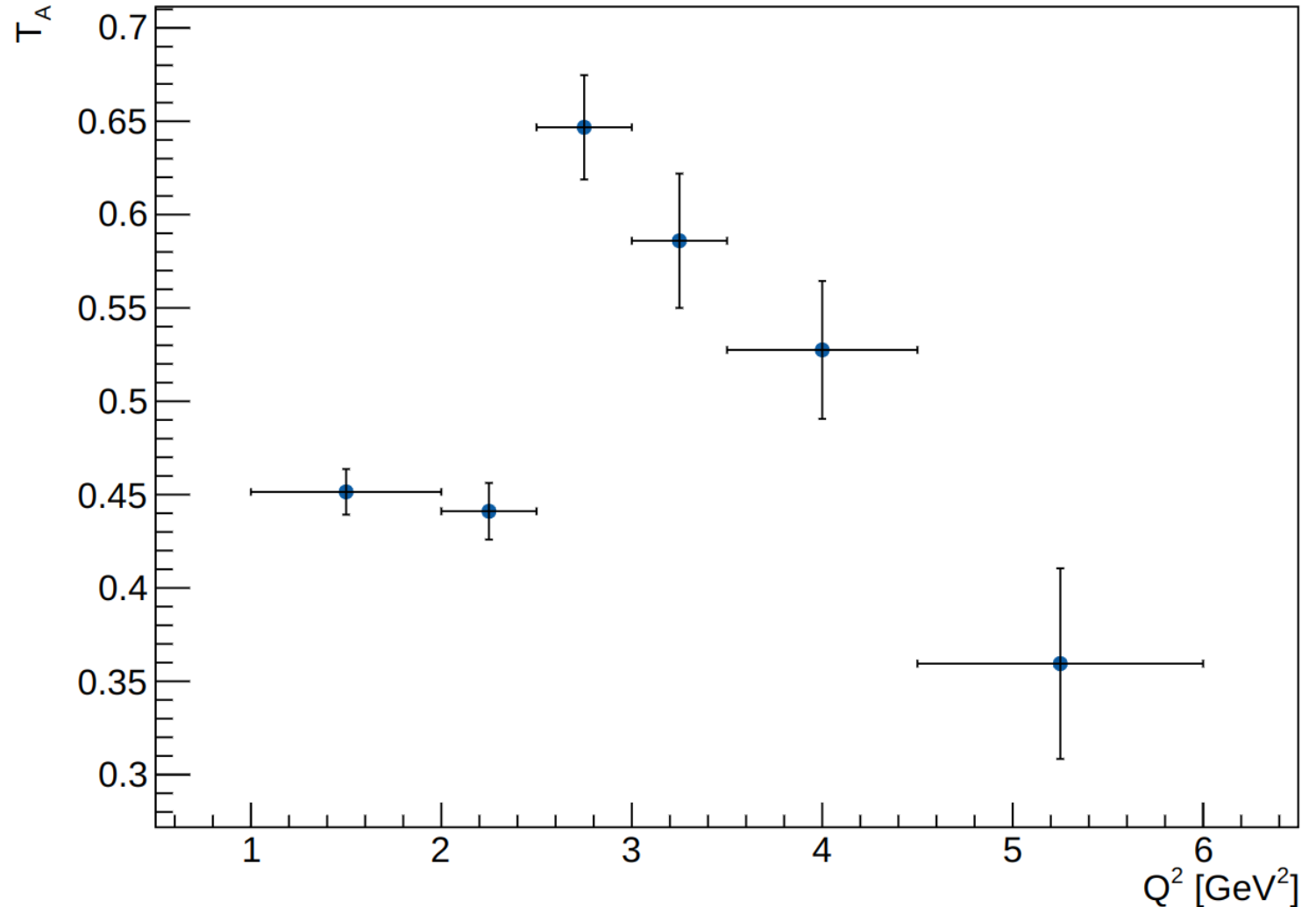
Nuclear transparency

- Take the integral between 3 sigma = 0.15:
- Expected to see no modification since we compute the integral between approximately 0.32 and 1.22 instead of 0.3 and 1.4.



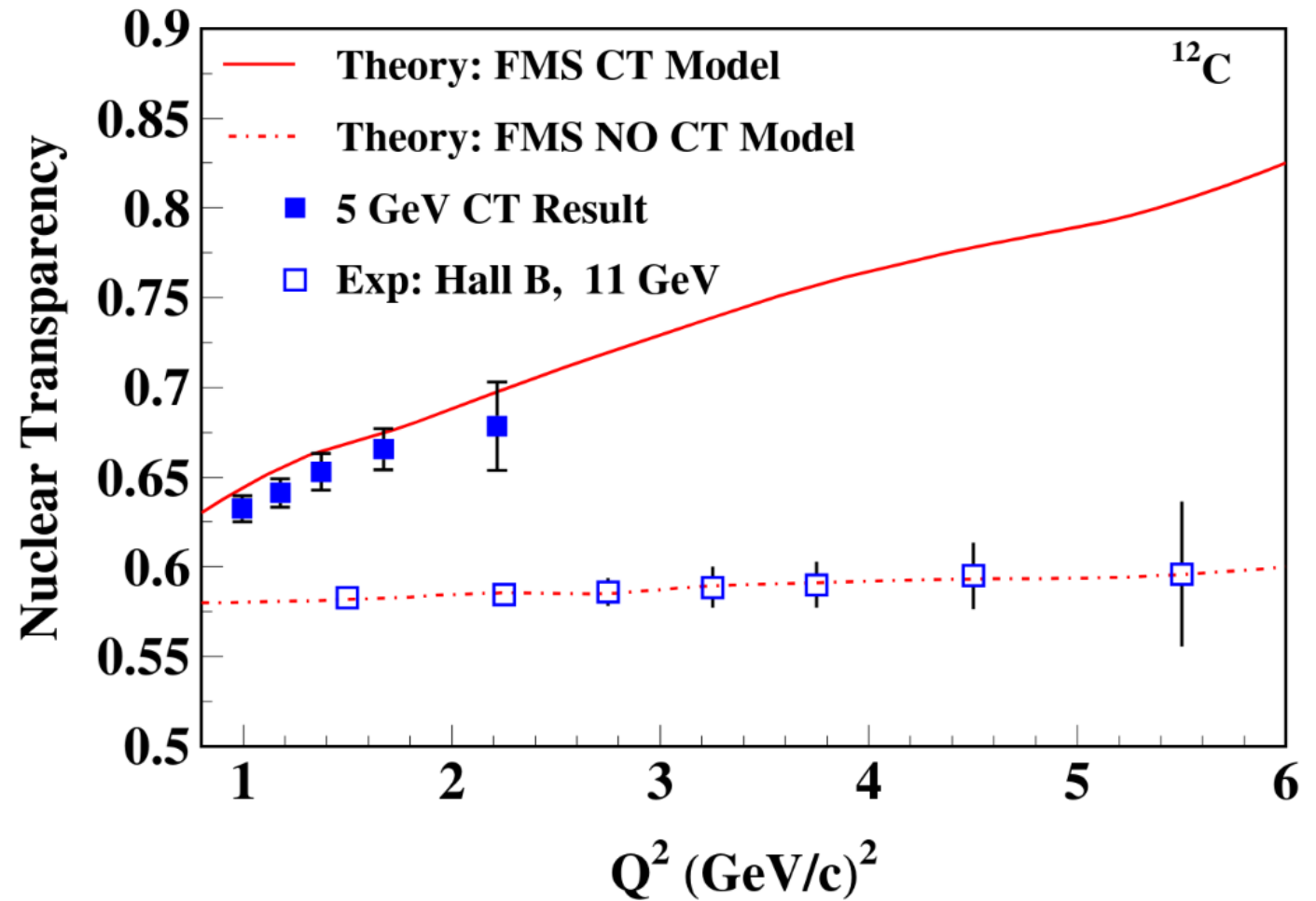
Nuclear transparency

- In the proposal the statistical errors are given for a bin in l_c between 0.4 and 0.5. Here is the nuclear transparency for this bin:
- Value and error for the last bin: $4.5 < Q^2 < 6 \text{ GeV}^2$ $T_A = 0.359 \pm 0.051$



Nuclear transparency

- Error in the proposal:
 $5 < Q^2 < 6$, $\text{error} = 0.015$
- Multiply the statistic by 10
reduce the error by $\sim 30\%$. So
 $30/100 * 0.05 = 0.015$



To do list:

- Cooking chef:
 - Clean files
- RG-D analysis:
 - Sector depend v_x and v_y + fit to cross check, compare to sector independant. (started)
 - SIDIS analysis, multiplicity ratio and cut on v_{txy} and v_{tz} and d_{oca} to see if it improve the peak, Δt vs p_{kaon}
 - Daniel cross check.
- AI ALERT:
 - training and testing with different particles (H, 2H, 3H, 3He, 4He).
 - MLP vs conventional (efficiency and speed)
- ALERT22GeV:
 - Look for low Q^2 , xB event
 - Generate events with double luminosity.
 - Compute asymmetry
 - Reproduce previous results

For the CLAS collaboration talk

- Should I make a slide for Suman analysis since he will present? No
- Should I include that at the end:
- This work is supported in part by the U.S. DOE award #: DE-FG02-07ER41528.
- At the end